

Unsupervised Learning in preparation for Pattern Recognition in k-Feldspar Using Cathodoluminescence Emission and EDX Composition Mapping



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1. INTRODUCTION

Feldspars are key minerals for luminescence dating (TL/OSL). Dating accuracy is challenged by anomalous fading and compositional variability.

Cathodoluminescence (CL) and Energy Dispersive X-ray Spectroscopy (EDX) enable spatially resolved insight into luminescent centers.

Objective: enrich raw CL/EDX data to build machine-learning features for unsupervised clustering and pattern discovery.

2. MATERIALS AND METHODS

Sample: R47 feldspar (Tanzania), grain size 120–300 μm .

Data: 92×128 pixels = 11,776 CL spectra (~1600 channels each) + 22 EDX elements (composition in Table 1).

Preprocessing: smoothing with Savitzky–Golay filter.

Peak extraction: 27,752 peaks detected $\rightarrow$ peak position, intensity, FWHM, area, barycenter.

Clustering: DBSCAN grouped peaks into emission domains (Fe^{3+} ~720 nm, IRCL ~880/955 nm).

Feature Engineering: 103 descriptors (peak metrics, ratios, coordinates, EDX composition).

3. RESULTS

Raw dataset: dominated by 1st PCA component (>85% variance) $\rightarrow$ high redundancy (Fig. 1A).

Enriched dataset: variance spread over 6 PCs (>80% variance) $\rightarrow$ richer structural diversity ((Fig 3A).

Clustering: DBSCAN separated spectral domains:

- Fe^{3+} bands (620–780 nm).
- IRCL 880 nm (Na-rich).
- IRCL 955 nm (K-rich).

Distinct emission-based domains linked to local chemical composition.

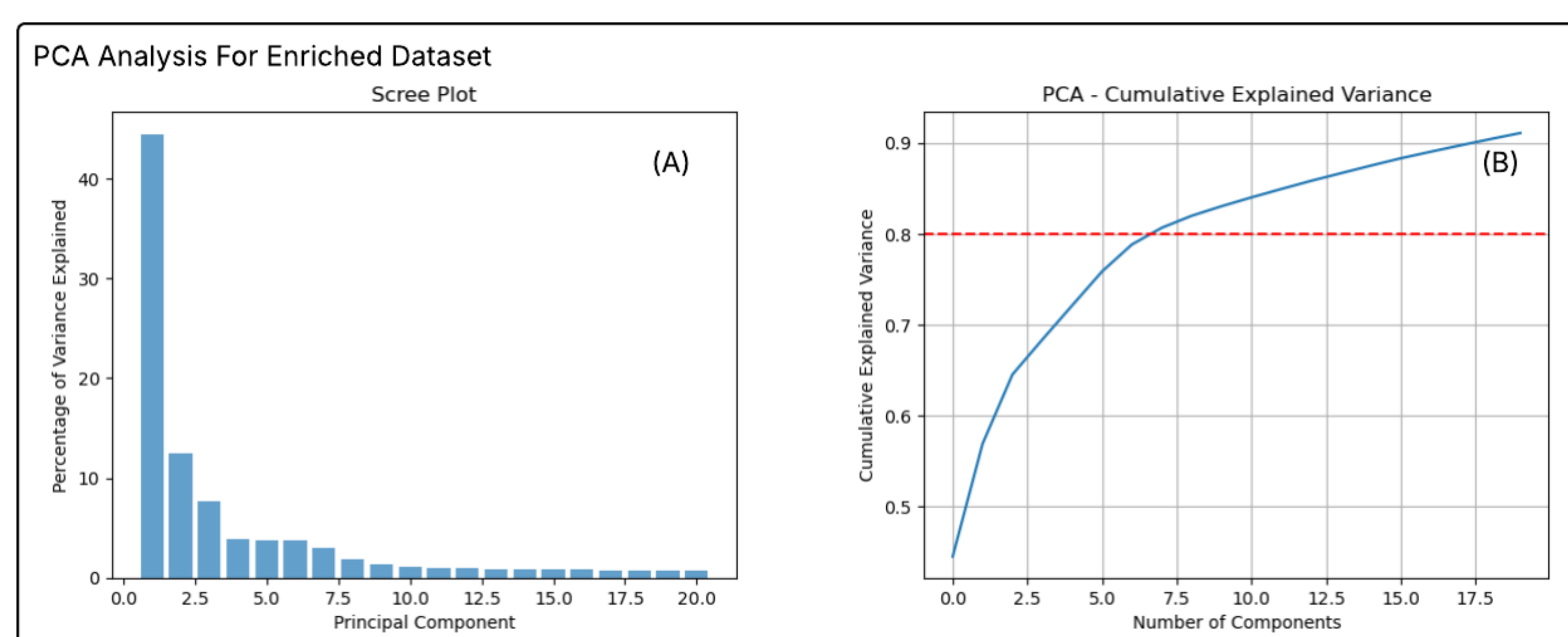
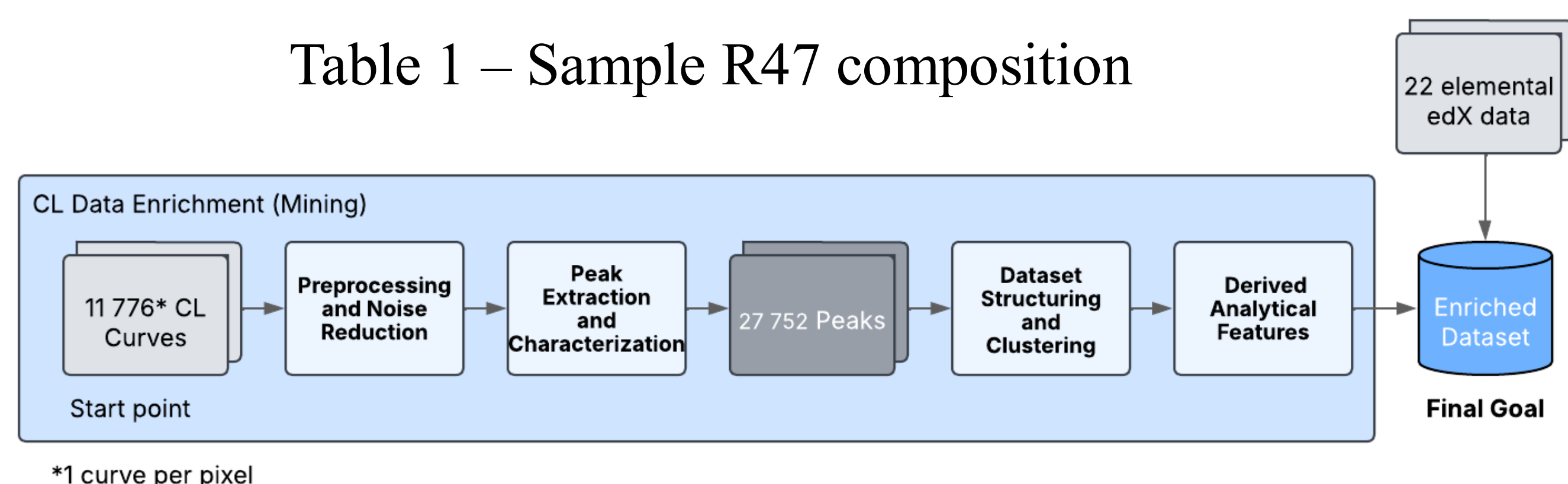


Figure 3 - Principal Component Analysis (PCA) of enriched dataset

Element	Atomic %
C	18.35 $\pm$ 0.12
O	53.62 $\pm$ 0.26
Na	0.36 $\pm$ 0.01
Mg	0.06 $\pm$ 0.01
Al	6.10 $\pm$ 0.03
Si	17.00 $\pm$ 0.06
K	4.43 $\pm$ 0.02
Ca	0.04 $\pm$ 0.01
Fe	0.02 $\pm$ 0.01
Ba	0.03 $\pm$ 0.01
Total	100.00

Table 1 – Sample R47 composition



*1 curve per pixel

Figure 1 – Data Enrichment Pipeline

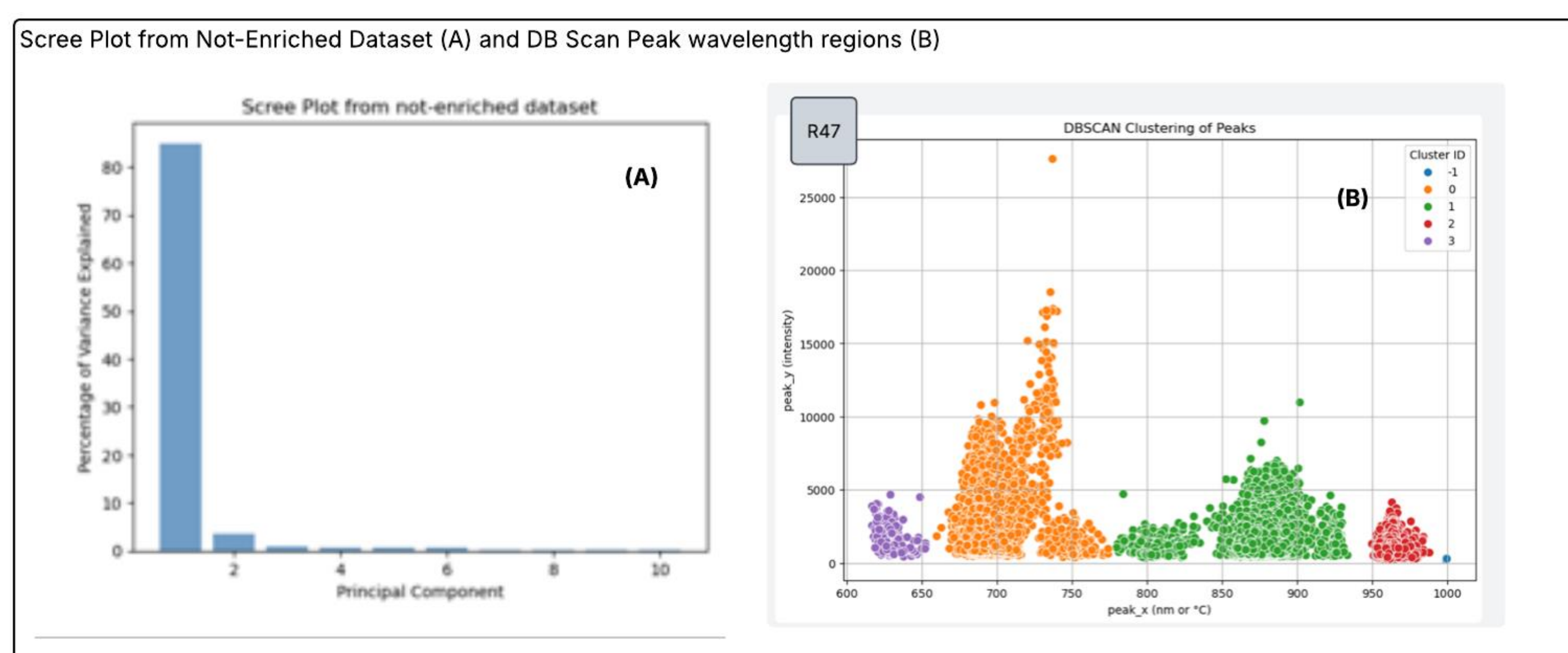


Figure 2 – PCA Scree Plot (A) and HDBSCAN Clustering by wavelength

4. CONCLUSIONS

Developed a scalable CL/EDX enrichment pipeline. Transformed redundant raw spectra into machine-learning-friendly datasets. PCA confirmed improved variance distribution and interpretability. DBSCAN clustering revealed luminescence–composition domains. Benchmark for mineralogical studies using unsupervised ML. Future Work: expand to other luminescent materials, kinetic/time-resolved CL, and deep learning segmentation.

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